NEET UG 2027: Chemistry Formula Compendium

Module 03: Inorganic Chemistry Master Compendium

Prepared, Compiled & Authored by Satya Prakash | **Subject:** Inorganic Chemistry (Units 3, 9, 10, 11 & 12)

Strict NCERT & NMC Alignment | Complete Mathematical Variations, Use-Cases & Traps

1 Classification of Elements & Periodicity (Unit 9)

1.1 Effective Nuclear Charge & Atomic Radii Trends

Slater's Rules, Screening Constant & Radius Scaling

$$Z_{\text{eff}} = Z - \sigma \qquad [Z = \text{Atomic Number}, \quad \sigma = \text{Screening / Shielding Constant}] \tag{1}$$

$$\text{Shielding Power Order} : s > p > d > f \qquad [f\text{- and } d\text{-electrons provide extraordinarily POOR shielding}] \tag{2}$$

$$r_{\text{cation}} < r_{\text{atom}} < r_{\text{anion}} \qquad [r_{\text{isoelectronic}} \propto \frac{1}{Z} \implies \text{Size: } O^{2-} > F^{-} > Na^{+} > Mg^{2+} > Al^{3+}] \tag{3}$$

1.2 Ionization Enthalpy, Electron Gain Enthalpy & Electronegativity

Periodic Property	General Trend Across Period / Down Group		High-Yield NEET Exceptions & Anomalies
Ionization Enthalpy (IE)	Across: Increases	Down: Decreases	Be > B ($2s^2$ fully filled vs $2p^1$); N > O ($2p^3$ half filled vs $2p^4$); Mg > Al; P > S; Tl > In (Lanthanoid contraction).
Electron Gain Enthalpy ($\Delta_{eg}H$)	Across: More Negative	Down: Less Negative	Cl > F and S > O! (Compact $2p$ shell in F and O creates intense inter-electronic repulsions, lowering incoming electron affinity).
Electronegativity (χ)	Across: Increases	Down: Decreases	Most electronegative: F(4.0) > O(3.5) > N(3.0) ≈ Cl(3.0) > Br(2.8) > I ≈ C ≈ S(2.5).

Table 1: Systematic Summary of Periodic Trends and Anomaly Anchors.

Electronegativity Scales & Dipole Moment Formulations

$$\chi_{\text{Mulliken}} = \frac{\text{IE} + \text{EA}}{2} \text{ (in eV)}, \qquad \chi_{\text{Pauling}} = \frac{\chi_{\text{Mulliken}}}{2.8} = \frac{\text{IE} + \text{EA}}{5.6} \tag{4}$$

$$\mu = q \times d \quad [1\,\text{Debye (D)} = 3.33564 \times 10^{-30}\,\text{C m}], \qquad \%\,\text{Ionic Character} = \frac{\mu_{\text{observed}}}{\mu_{\text{calculated}}} \times 100\% \tag{5}$$

$$\mu_{\text{net}} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta} \implies \mu_{\text{ortho}} > \mu_{\text{meta}} > \mu_{\text{para}} = 0 \quad \text{(for identical substituents)} \tag{6}$$

Periodic Traps: NH_3 vs NF_3 Dipole Moment & Electron Gain Enthalpy

- **Dipole Moment Comparison ($\mu_{NH_3} > \mu_{NF_3}$):** In NH_3, bond dipoles of three N – H bonds reinforce the lone-pair dipole ($\mu = 1.47\,\text{D}$). In NF_3, the highly electronegative fluorines pull electron density away, opposing the lone-pair dipole ($\mu = 0.23\,\text{D}$).
- **Most Negative $\Delta_{eg}H$:** The element with the **most negative electron gain enthalpy** in the entire periodic table is **Chlorine (Cl)**, NOT Fluorine ($\text{Cl} : -349\,\text{kJ mol}^{-1}$ vs $\text{F} : -328\,\text{kJ mol}^{-1}$).

2 Chemical Bonding & Molecular Structure (Unit 3)

2.1 Formal Charge & VSEPR Theory Master Geometry Table

Formal Charge Equation & Steric Number Rule

$$\text{Formal Charge (FC)} = V - L - \frac{1}{2}S \qquad [V = \text{Valence} \tag{7}$$

$$\text{Steric Number (SN)} = (\text{Number of } \sigma\text{-bonds}) + (\text{Number of localized lone pairs on central atom}) \tag{8}$$

SN	Hybridization	Bond Pairs	Lone Pairs	Molecular Geometry / Shape	High-Yield NEET Examples
2	sp	2	0	Linear (180°)	$BeCl_2, CO_2, HCN, C_2H_2$
3	sp^2	3	0	Trigonal Planar (120°)	$BF_3, BCl_3, SO_3, NO_3^-$
3	sp^2	2	1	Bent / V-Shaped ($< 120°$)	SO_2, O_3, NO_2^-
4	sp^3	4	0	Tetrahedral (109.5°)	$CH_4, CCl_4, NH_4^+, SO_4^{2-}$
4	sp^3	3	1	Trigonal Pyramidal (107°)	$NH_3, PCl_3, H_3O^+, ClO_3^-$
4	sp^3	2	2	Bent / Angular (104.5°)	H_2O, H_2S, OF_2, SCl_2
5	sp^3d	5	0	Trigonal Bipyramidal	$PCl_5, PF_5, SbCl_5$
5	sp^3d	4	1	Seesaw / Distorted Tetrahedral	SF_4, SeF_4
5	sp^3d	3	2	T-Shaped	ClF_3, BrF_3, ICl_3
5	sp^3d	2	3	Linear	XeF_2, I_3^-, ICl_2^-
6	sp^3d^2	6	0	Octahedral (90°)	$SF_6, [AlF_6]^{3-}, PF_6^-$
6	sp^3d^2	5	1	Square Pyramidal	$BrF_5, IF_5, XeOF_4$
6	sp^3d^2	4	2	Square Planar	$XeF_4, [ICl_4]^-$
7	sp^3d^3	7	0	Pentagonal Bipyramidal	IF_7
7	sp^3d^3	6	1	Distorted Octahedral	XeF_6

Table 2: Master VSEPR Geometries, Hybridization and Lone Pair Partitions.

Bent's Rule Formulations in Trigonal Bipyramidal (sp^3d)

- **Bent's Postulate:** More electronegative substituents prefer hybrid orbitals having **less** s**-character** (Axial bonds in TBP: pure pd character, longer and weaker).
- **Lone Pair Preference:** Lone pairs and electropositive substituents prefer hybrid orbitals having **more** s**-character** (Equatorial bonds in TBP: sp^2 character, shorter and stronger).
- **Application to PCl_5 and ClF_3:** In gaseous PCl_5, axial $P-Cl$ bonds (214 pm) are longer than equatorial bonds (202 pm). In solid state, it dissociates into ionic $[PCl_4]^+[sp^3]$ and $[PCl_6]^-[sp^3d^2]$.

2.2 Molecular Orbital Theory (MOT)

Molecular Orbital Energy Levels, Bond Order & Magnetic State

$$\text{Bond Order (BO)} = \frac{1}{2}(N_b - N_a), \qquad \text{Stability} \propto \text{BO} \propto \text{Bond Dissociation Energy} \propto \frac{1}{\text{Bond Length}} \tag{9}$$

Case 1: Total Electrons ≤ 14 (B_2, C_2, N_2 with sp-mixing):

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z \tag{10}$$

Case 2: Total Electrons > 14 (O_2, F_2 without sp-mixing):

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z \tag{11}$$

Species	Total Electrons	Configuration Key	Bond Order	Magnetic Behaviour
N_2	14	$(\pi 2p)^4(\sigma 2p_z)^2$	3.0	Diamagnetic
N_2^+	13	$(\pi 2p)^4(\sigma 2p_z)^1$	2.5	Paramagnetic (1 unpaired e^-)
O_2^{2+}	14	$(\sigma 2p_z)^2(\pi 2p)^4$	3.0	Diamagnetic
O_2^+	15	$(\pi^* 2p)^1$	2.5	Paramagnetic (1 unpaired e^-)
O_2	16	$(\pi^* 2p_x)^1(\pi^* 2p_y)^1$	2.0	Paramagnetic (2 unpaired e^-)
O_2^- (Superoxide)	17	$(\pi^* 2p)^3$	1.5	Paramagnetic (1 unpaired e^-)
O_2^{2-} (Peroxide)	18	$(\pi^* 2p)^4$	1.0	Diamagnetic
C_2	12	$(\pi 2p)^4$	2.0	Diamagnetic (Double bond consists of **two π-bonds**!)

Table 3: Molecular Orbital Configurations, Bond Orders and Paramagnetism for Homonuclear Diatomics.

3 Coordination Compounds (Unit 12)

3.1 Werner's Theory, Ligands & Chelate Effect

Primary vs Secondary Valency & Chelate Stability

$$\text{Primary Valency} = \text{Oxidation state of metal; ionizable; non-directional} \tag{12}$$

$$\text{Secondary Valency} = \text{Coordination number of metal; non-ionizable; directional (governs geometry)} \tag{13}$$

$$\text{Chelate Effect : Polydentate ligands form stable ring structures} \implies \Delta S^\circ > 0 \implies \Delta G^\circ < 0 \tag{14}$$

3.2 Crystal Field Theory (CFT) & Splitting Formulations

Octahedral & Tetrahedral Crystal Field Splitting Parameters

$$\Delta_t = \frac{4}{9}\Delta_o \implies \Delta_t < P \quad \text{(Tetrahedral complexes are ALWAYS high spin!)} \tag{15}$$

$$\text{CFSE}_{\text{octahedral}} = \left(-0.4\, n_{t_{2g}} + 0.6\, n_{e_g}\right)\Delta_o + mP \tag{16}$$

$$\text{Spectrochemical Series : } \mathrm{I^- < Br^- < S^{2-} < SCN^- < Cl^- < N_3^- < F^- < OH^- < C_2O_4^{2-} \approx H_2O} \tag{17}$$

$$\mathrm{< NCS^- < EDTA^{4-} < NH_3 \approx py < en < bpy < phen < NO_2^- < PPh_3 < CN^- \approx CO} \tag{18}$$

High-Spin vs Low-Spin Splitting Conditions for Octahedral Complexes:

$$\Delta_o > P \implies \text{Strong field ligand; pairing occurs in } t_{2g} \implies \text{Low-Spin / Inner Orbital Complex} \tag{19}$$

$$\Delta_o < P \implies \text{Weak field ligand; electrons occupy } e_g \text{ before pairing} \implies \text{High-Spin / Outer Orbital Complex} \tag{20}$$

Ion Configuration	Weak Field ($\Delta_o < P$)	CFSE (Weak)	Strong Field ($\Delta_o > P$)	CFSE (Strong)
d^4 (Cr^{2+}, Mn^{3+})	$t_{2g}^3 e_g^1$ ($n=4$)	$-0.6\,\Delta_o$	$t_{2g}^4 e_g^0$ ($n=2$)	$-1.6\,\Delta_o + P$
d^5 (Fe^{3+}, Mn^{2+})	$t_{2g}^3 e_g^2$ ($n=5$)	$0\,\Delta_o$	$t_{2g}^5 e_g^0$ ($n=1$)	$-2.0\,\Delta_o + 2P$
d^6 (Fe^{2+}, Co^{3+})	$t_{2g}^4 e_g^2$ ($n=4$)	$-0.4\,\Delta_o$	$t_{2g}^6 e_g^0$ ($n=0$)	$-2.4\,\Delta_o + 2P$
d^7 (Co^{2+})	$t_{2g}^5 e_g^2$ ($n=3$)	$-0.8\,\Delta_o$	$t_{2g}^6 e_g^1$ ($n=1$)	$-1.8\,\Delta_o + P$

Table 4: Crystal Field Electronic Configurations and CFSE Formulations for $3d^4 - 3d^7$ Octahedral Ions.

3.3 Stereoisomerism in Coordination Complexes

Geometrical and Optical Isomerism Rules

- **Square Planar Complexes ($CN = 4$):**
  - $[MA_2B_2]$: Exhibits 2 geometrical isomers (cis and trans).
  - $[MABCD]$: Exhibits 3 geometrical isomers (2 cis, 1 trans).
  - Square planar complexes **never exhibit optical isomerism** because the molecular plane is an inherent plane of symmetry (σ).
- **Octahedral Complexes ($CN = 6$):**
  - $[MA_4B_2]$: Exhibits 2 geometrical isomers (cis and trans). Both are optically inactive.
  - $[MA_3B_3]$: Exhibits 2 geometrical isomers (facial *fac* and meridional *mer*). Both are optically inactive.
  - $[M(AA)_3]$ (e.g. $[Co(en)_3]^{3+}$): Exhibits 0 geometrical, but exists as a pair of **optical enantiomers** (d and l).
  - $[M(AA)_2B_2]$ (e.g. $[Co(en)_2Cl_2]^{+}$): *cis*-isomer is chiral and optically active (d/l pair); *trans*-isomer has plane of symmetry and is **optically inactive**.

4 The d- and f-Block Elements (Unit 11)

4.1 Transition Metal Trends & Variable Oxidation States

Magnetic Moment, Enthalpy of Atomization & Electrode Potentials

$$\mu_{\text{spin-only}} = \sqrt{n(n+2)}\,\text{BM} \tag{21}$$

$$\text{Max Oxidation State} = \text{Sc } (+3),\quad \text{Ti } (+4),\quad \text{V } (+5),\quad \text{Cr } (+6),\quad \text{Mn } (+7),\quad \text{Fe } (+6),\quad \text{Os/Ru } (+8 \text{ in } OsO_4) \tag{22}$$

4.2 Potassium Dichromate ($K_2Cr_2O_7$) & Permanganate ($KMnO_4$)

Reagent & Reaction Medium	Change in Oxidation State	n-Factor	Canonical Half-Reaction
$K_2Cr_2O_7$ in Acidic Medium	$Cr^{6+} \to Cr^{3+}$	6	$Cr_2O_7^{2-} + 14H^+ + 6e^- \to 2Cr^{3+} + 7H_2O$
$KMnO_4$ in Acidic Medium	$Mn^{7+} \to Mn^{2+}$	5	$MnO_4^- + 8H^+ + 5e^- \to Mn^{2+} + 4H_2O$
$KMnO_4$ in Neutral / Faintly Alkaline	$Mn^{7+} \to Mn^{4+}$	3	$MnO_4^- + 2H_2O + 3e^- \to MnO_2 + 4OH^-$
$KMnO_4$ in Strongly Alkaline Medium	$Mn^{7+} \to Mn^{6+}$	1	$MnO_4^- + e^- \to MnO_4^{2-}$

Table 5: Stoichiometric Redox n-Factors for Dichromate and Permanganate Titrations.

High-Yield Oxidation Reactions of $KMnO_4$ and $K_2Cr_2O_7$

- **Acidic $KMnO_4$ converts:**

$$Fe^{2+} \to Fe^{3+},\quad C_2O_4^{2-} \to CO_2,\quad S^{2-} \to S,\quad SO_3^{2-} \to SO_4^{2-},\quad NO_2^- \to NO_3^-,\quad I^- \to I_2$$

- **Faintly Alkaline $KMnO_4$ (Baeyer's Reagent) converts:**

$$I^- \to IO_3^- \text{ (Iodate ion, NOT } I_2\text{!)},\qquad S_2O_3^{2-} \to SO_4^{2-} \text{ (Thiosulfate to Sulfate)}$$

- **Chromyl Chloride Test for Chloride (Cl^-) Detection:**

$$K_2Cr_2O_7 + 4NaCl + 6H_2SO_4 \xrightarrow{\Delta} 2CrO_2Cl_2\uparrow \text{ (Deep Red Vapour)} + 2KHSO_4 + 4NaHSO_4 + 3H_2O$$

4.3 Lanthanoid Contraction & Oxidation States

Lanthanoid Contraction Governing Principles

$$\text{Radius Trend}: \text{La}^{3+}(106\,\text{pm}) \rightarrow \text{Lu}^{3+}(86\,\text{pm}) \quad [\text{Steady contraction of } 20\,\text{pm across } 14 \text{ elements}] \tag{23}$$

$$\text{Radius Similarity}: \text{Zr}(160\,\text{pm}) \approx \text{Hf}(159\,\text{pm}), \quad \text{Nb}(146\,\text{pm}) \approx \text{Ta}(146\,\text{pm}), \quad \text{Mo} \approx \text{W} \tag{24}$$

$$\text{Basicity Order}: \text{La(OH)}_3 > \text{Ce(OH)}_3 > \cdots > \text{Lu(OH)}_3 \quad [\text{Covalent character increases as cation size shrinks}] \tag{25}$$

High-Yield Oxidation States of Lanthanoids:

$$\text{Dominant: } +3; \quad \text{Ce}^{4+} \ ([\text{Xe}]\ 4f^0 \text{ powerful oxidizing agent}), \quad \text{Eu}^{2+} \ ([\text{Xe}]\ 4f^7 \text{ reducing agent}), \quad \text{Yb}^{2+} \ ([\tag{26}$$

5 p-Block Elements (Unit 10)

5.1 Inert Pair Effect & Thermodynamic Stability Trends

Group	Higher Oxidation State	Lower Oxidation State	Stability Order & Redox Consequence
Group 13 (B, Al, Ga, In, Tl)	+3	+1	$Tl^{+} > Tl^{3+} \implies$ Tl^{3+} is oxidizing; Al^{3+} is stable.
Group 14 (C, Si, Ge, Sn, Pb)	+4	+2	$Pb^{2+} > Pb^{4+} \implies$ PbO_2 is powerful oxidizer; Sn^{2+} is red
Group 15 (N, P, As, Sb, Bi)	+5	+3	$Bi^{3+} > Bi^{5+} \implies$ BiF_5 only stable pentahalide; Bi^{5+} ox

Table 6: Inert Pair Effect Manifestation Across Heavy p-Block Elements.

Group 15, 16 & 17 Hydrides High-Yield Periodic Trends

- **Group 15 Hydrides ($NH_3, PH_3, AsH_3, SbH_3, BiH_3$):**
  - Bond Angle: $NH_3(107.8°) > PH_3(93.6°) > AsH_3(91.8°) > SbH_3(91.3°)$ (Drago's rule).
  - Thermal Stability: $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$.
  - Reducing Character: $BiH_3 > SbH_3 > AsH_3 > PH_3 > NH_3$ (strongest reducing agent is BiH_3).
  - Boiling Point: $BiH_3 > SbH_3 > \mathbf{NH_3} > AsH_3 > PH_3$ (NH_3 elevated due to H-bonding!).
- **Group 16 Hydrides (H_2O, H_2S, H_2Se, H_2Te):**
  - Acidic Strength: $H_2O < H_2S < H_2Se < H_2Te$ ($E-H$ bond enthalpy decreases).
  - Boiling Point: $\mathbf{H_2}O > H_2Te > H_2Se > H_2S$ (water is highest due to extensive intermolecular H-bonding).
- **Group 17 Hydrides (HF, HCl, HBr, HI):**
  - Acidic Strength: $HF < HCl < HBr < HI$ (HI is strongest acid).
  - Boiling Point: $HF > HI > HBr > HCl$ (HF elevated due to Zig-Zag H-bonding).
  - Reducing Power: $HI > HBr > HCl > HF$.

p-Block Traps: Oxoacids of Phosphorus & Boric Acid Basicity

- **Basicity and Reducing Nature of Phosphorus Oxoacids:**
  - H_3PO_2 (Hypophosphorous / Phosphinic acid): Contains $1\,P-OH$ and $2\,P-H$ bonds. **Monobasic**; powerful reducing agent.
  - H_3PO_3 (Phosphorous / Phosphonic acid): Contains $2\,P-OH$ and $1\,P-H$ bonds. **Dibasic**; moderate reducing agent.
  - H_3PO_4 (Orthophosphoric acid): Contains $3\,P-OH$ and $0\,P-H$ bonds. **Tribasic**; non-reducing.
- **Orthoboric Acid (H_3BO_3):** Is **not a protonic acid**! It is a weak, **monobasic Lewis acid** that accepts OH^- from water:

$$B(OH)_3 + 2H_2O \rightleftharpoons [B(OH)_4]^- + H_3O^+$$

End of Inorganic Chemistry Master Compendium